

MySQL Capstone Project 2

[illegible]

Business Scenario

A hospital wants to analyze:

- Patients
- Doctors
- Departments
- Appointments
- Treatments
- Bills
- Payments

The goal is to build a MySQL database and generate reports for hospital management.

Tables Required

patients
doctors
departments
appointments
treatments
bills
payments

Minimum Records

Table	Minimum Records
patients	12
doctors	8
departments	5
appointments	20
treatments	15
bills	15
payments	15

Required Tables

patients

```
CREATE TABLE patients
(
    patient_id INT PRIMARY KEY,
    patient_name VARCHAR(100),
    gender VARCHAR(10),
    age INT,
    city VARCHAR(50),
    phone VARCHAR(15)
);
```

departments

```
CREATE TABLE departments
(
    department_id INT PRIMARY KEY,
    department_name VARCHAR(100)
);
```

doctors

```
CREATE TABLE doctors
(
    doctor_id INT PRIMARY KEY,
    doctor_name VARCHAR(100),
    specialization VARCHAR(100),
    department_id INT,
    consultation_fee DECIMAL(10,2)
);
```

appointments

```
CREATE TABLE appointments
(
    appointment_id INT PRIMARY KEY,
    patient_id INT,
    doctor_id INT,
    appointment_date DATE,
    appointment_status VARCHAR(30)
);
```

treatments

```
CREATE TABLE treatments
(
    treatment_id INT PRIMARY KEY,
    appointment_id INT,
    treatment_name VARCHAR(100),
    treatment_cost DECIMAL(10,2)
);
```

bills

```
CREATE TABLE bills
(
    bill_id INT PRIMARY KEY,
    patient_id INT,
    appointment_id INT,
```

```
bill_date DATE,  
total_amount DECIMAL(10,2),  
bill_status VARCHAR(30)  
);
```

payments

```
CREATE TABLE payments  
(  
    payment_id INT PRIMARY KEY,  
    bill_id INT,  
    payment_mode VARCHAR(30),  
    paid_amount DECIMAL(10,2),  
    payment_status VARCHAR(30)  
);
```

Capstone Tasks

Part 1: Basic Queries

1. Display all patients.
 2. Display all doctors.
 3. Find patients from Hyderabad.
 4. Find doctors from Cardiology department.
 5. Find appointments scheduled after 2026-01-01 .
 6. Find cancelled appointments.
 7. Find bills where total amount is greater than ₹5,000.
 8. Find payments made using UPI.
 9. Display patients aged between 30 and 50.
 10. Display doctors with consultation fee above ₹800.
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Part 2: Aggregate Queries

11. Count total patients.
12. Count total doctors.

13. Count total appointments.
 14. Find average consultation fee.
 15. Find highest treatment cost.
 16. Find total billing amount.
 17. Find total paid amount.
 18. Count patients by city.
 19. Count doctors by specialization.
 20. Count appointments by status.
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Part 3: Joins

21. Display patient name with appointment date and status.
22. Display doctor name with department name.
23. Display patient name, doctor name, and appointment date.
24. Display appointment ID with treatment name and cost.
25. Display bill ID with patient name and total amount.
26. Display bill ID with payment mode, paid amount, and payment status.
27. Create a full appointment report:

```
Patient Name
Doctor Name
Department
Appointment Date
Appointment Status
Treatment Name
Treatment Cost
Bill Amount
Payment Status
```

Part 4: GROUP BY and HAVING

28. Count appointments by doctor.
29. Count appointments by department.
30. Total revenue by department.
31. Total treatment cost by treatment name.
32. Total billing by city.
33. Doctors having more than 2 appointments.
34. Departments generating revenue greater than ₹20,000.
35. Cities having more than 2 patients.

Part 5: Subqueries

- 36. Find patients who have appointments.
 - 37. Find patients who never booked appointments.
 - 38. Find doctors who have no appointments.
 - 39. Find bills greater than average bill amount.
 - 40. Find patient with highest bill amount.
 - 41. Find doctors whose consultation fee is above average.
 - 42. Find patients who visited Cardiology.
 - 43. Find unpaid bills.
 - 44. Find appointments that have treatments.
 - 45. Find patients whose total bill is above average patient billing.
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Part 6: Data Quality Checks

- 46. Find appointments without treatment.
 - 47. Find bills without payment.
 - 48. Find payments with NULL or zero paid amount.
 - 49. Find cancelled appointments that still have bills.
 - 50. Find paid bills where payment amount is less than bill amount.
 - 51. Find doctors with invalid department ID.
 - 52. Find appointments with invalid patient or doctor IDs.
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Final Reports

Report 1: Patient Billing Report

```
Patient Name
City
Total Appointments
Total Bill Amount
Total Paid Amount
Pending Amount
```